

Geometric Shapes: Areas for Puzzle Design

Name: _____

Class: 6th Grade Math

This worksheet will help you review the area formulas for rectangles, triangles, and circles. Use these to plan the shapes in your 15 cm x 20 cm jigsaw puzzle design. Remember, the total puzzle area is 15 cm x 20 cm = 300 cm². Calculate areas to ensure your shapes fit!

Section 1: Rectangles

Formula: Area = length x width

(Example: A rectangle that is 5 cm long and 3 cm wide has an area of $5 \times 3 = 15$ cm².)

1. Draw a rectangle on the grid with length 4 cm and width 2 cm. Calculate its area.

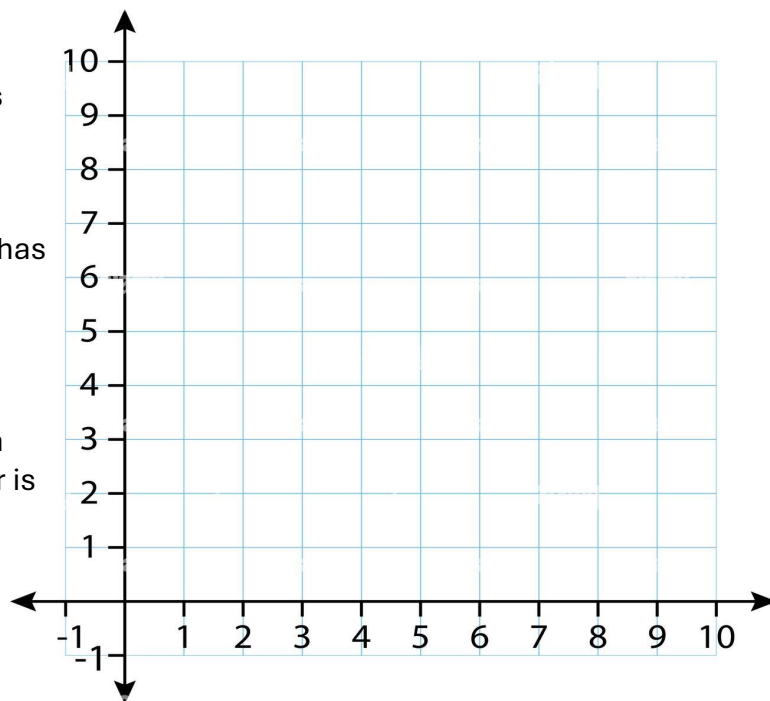
Area = ____ cm²

2. If a rectangle in your puzzle design has a length of 10 cm and an area of 50 cm², what is the width?

Width = ____ cm

3. Compose two rectangles to make a larger shape: One is 6 cm x 4 cm, the other is 3 cm x 4 cm. What is the total area?

Total Area = ____ cm²



Section 2: Triangles

(Example: A triangle with base 6 cm and height 4 cm has an area of $(6 \times 4) \div 2 = 12$ cm².)

Formula: Area = (base x height) ÷ 2

4. Draw a triangle on the grid above with base 5 cm and height 3 cm. Calculate its area.

Area = ____ cm²

Section 3: Circles

Formula: $= \pi r^2$ (Use 3.14 for pi)

(Example: A circle with radius 2 cm has an area of $3.14 \times 2 \times 2 = 12.56$ cm².)

5. Draw a circle on the grid below with radius 3 cm. Calculate its area.

Area = ____ cm² (round to two decimals) You can show your work below: 3.14 x 3 x 3